

Unraveling the Linux kernel

A look at the Linux kernel internals, the versioning system and how it affects us as end-users

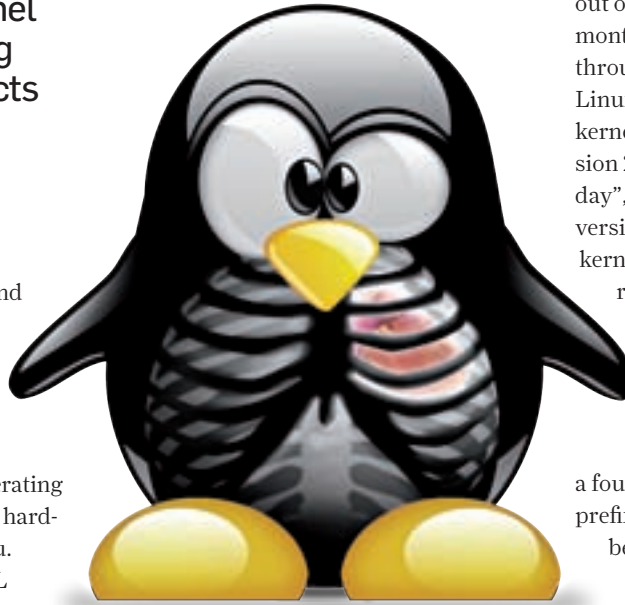
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Linux kernel is the central and most important part of the Linux operating system, not to mention the component that has given the system its name. In the most simplest of terms, a kernel is that part of the operating system which sits directly above the hardware doing all the dirty work for you. The kernel along with the GNU/GPL Software Suite is what gives us the operating system as we know today.

Supposed to be the next best thing since sliced bread, it all started when Linus Torvalds, a student at the University of Helsinki got the idea of making a task switcher with a terminal driver for an Intel 80386. After months of work, he came up with something that loosely resembled an operating system but was limited in application to a 386. Now this was a time when Usenet groups were still considered to be hip. Here he posted the now famous message, "I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386 (486) AT clones. This has been brewing since april...(sic)" and the developers flocked around like flies on dung. The rest, as they say, is history.

The kernel contains all the juiciest parts that you'd need to run an operating system – a process scheduler, a memory management unit, file system, device drivers, low-level drivers (for CPU, MMU and other onboard devices), inter-process communication system, I/O scheduler, I/O system and network services. As you can see, even drivers are a part of the kernel (which in case of Windows, most of them are not a part of the kernel and run in the user space). Infact fifty per cent of all the lines in the kernel source code are drivers,



thirty per cent are architecture-specific stuff. The processor, network and the core kernel code is only five per cent of the overall code. According to one calculation made by the current maintainer of the stable branch, 4,945 lines were added, 2,006 were removed and 1,702 lines were modified per day. As of 2012, the Linux 3.2 release had 14,998,651 lines of code.

This brings us to the current development model of the kernel. Linus Torvalds releases the versions which contain the main, generic branch of development, called the "vanilla" or "mainline" kernels. This branch is released as a new version after Linus assimilates the changes made by other developers in the code and after a round of pre-release bug fixes.

In the current lay of the land, the vanilla branch is not the "stable" branch, instead it incorporates all the changes including the bug and security fixes. For those who don't want to experiment with the cutting-edge version, there is the "long-term stable" branch which just adds the required bug and security fixes and not any changes to the kernel.

Since 2004, the release process changed and new kernels started coming

out on a regular schedule every 2-3 months, numbered 2.6.0, 2.6.1, right through to 2.6.39. After this, to mark Linux kernel's 20th anniversary the kernel number was bumped up from version 2.6 to 3.0. "Gone are the 2.6.<bignum> day", announced Linus. The current stable version is 3.2.7, while the latest mainline kernel is 3.3-rc4. The first two numbers

represent major revisions in the code while the last number represents bug, security fix increments.

Vastly different from the 2.6.x versioning system where x was incremented with each update and a fourth number was used for security prefixes. The old even-odd version numbering system is no longer in use.

Do these numbers affect you in any way? Well, most of the time, no. Most of the additions are to the "drivers" part of the code, while the updates to the architecture and the core, though substantial, are not well tested. Therefore, you should use the mainline kernel if you want to run the latest toaster that you bought yesterday or the fastest processor available on the planet. The distro that you are using may ship the "vanilla" and/or "stable" kernels. However, several Linux distribution vendors (such as Red Hat and Debian) maintain another set of Linux kernel branches which are integrated into their products. These are by and large updated at a slower pace compared to the "vanilla" branch, and they usually include all fixes from the relevant "stable" branch.

Which kernel are you using? You can find out the version you are using by entering `uname -r` in your terminal. How to update? Most of the distributions nowadays come with package managers and update utilities. So as soon as your distro's repositories get the image of the latest kernel, the changes will be pushed to your system. You can also use the release candidate, but you'll have to compile it yourself. If you're unsure how to go about it, instructions for doing that are at <http://goo.gl/NhKJu> and <http://goo.gl/Pv4zc>. 